

Physical Chemistry Practical: Topic 5 – Chemical Energetics

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Enthalpy of Displacement of Copper(II) by Zinc with Cooling Curve Extrapolation

Cambridge Syllabus Reference: 9701/31/M/J/24/Q2 · [10 Marks]

Reaction: $\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$

Method: 50.0 cm^3 of $0.800 \text{ mol dm}^{-3}$ CuSO_4 is placed in a polystyrene cup. Temperature is recorded every 30 seconds. At $t = 3.0 \text{ min}$, 3.00 g zinc powder is added (excess). Readings continue until $t = 10.0 \text{ min}$.

(a) Complete the thermometry data table below. [3]

Time / min	Temp / °C	Time / min	Temp / °C	Time / min	Temp / °C
0.0		3.5		7.0	
0.5		4.0		7.5	
1.0		4.5		8.0	
1.5		5.0		8.5	
2.0		5.5		9.0	
2.5		6.0		9.5	
3.0 (mix)	X	6.5		10.0	

(b) Explain why the temperature is extrapolated to $t = 3.0 \text{ min}$ rather than taking the maximum observed temperature. [2]

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(c) If $\Delta T = 24.5^\circ\text{C}$, calculate the heat energy released, q . [$c = 4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$, density = 1.00 g cm^{-3}] [2]

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(d) Calculate the molar enthalpy of displacement, ΔH_{disp} , in kJ mol^{-1} . [3]

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Question 2: Determination of Enthalpy Change of Neutralization (HCl + NaOH)**Cambridge Syllabus Reference: 9701/33/O/N/21/Q2 · [10 Marks]**

Method: 50.0 cm³ of 2.00 mol dm⁻³ HCl is mixed with 50.0 cm³ of 2.00 mol dm⁻³ NaOH in a polystyrene cup. Initial temperatures: HCl = 20.5°C, NaOH = 20.5°C. Maximum temperature reached = 33.8°C.

(a) Calculate the heat evolved in this neutralization reaction. [Assume density = 1.00 g cm⁻³, c = 4.18 J g⁻¹ °C⁻¹]
[3]

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(b) Calculate the amount, in moles, of water formed. [2]

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(c) Calculate the molar enthalpy of neutralization, ΔH_{neut} . [2]

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(d) State why ΔH_{neut} for all strong acid-strong base reactions is approximately -57 kJ mol⁻¹. [3]

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Question 3: Comparison of Neutralization Enthalpies: Ethanoic Acid vs Hydrochloric Acid

Cambridge Syllabus Reference: 9701/34/O/N/23/Q2 · [10 Marks]

When 50.0 cm^3 of 2.00 mol dm^{-3} CH_3COOH is neutralized by 50.0 cm^3 of 2.00 mol dm^{-3} NaOH , the temperature rise is 12.1°C (yielding $\Delta H = -50.6 \text{ kJ mol}^{-1}$), which is significantly less exothermic than with HCl .

(a) Explain why ethanoic acid produces a less exothermic enthalpy of neutralization than hydrochloric acid. [4]

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(b) Write the ionic equation for the neutralization of ethanoic acid with aqueous hydroxide ions. [2]

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(c) Suggest why a student using a glass beaker instead of a polystyrene cup obtained an even lower temperature rise. [2]

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(d) Calculate the percentage difference between the experimental ΔH for ethanoic acid (-50.6) and strong acid ($-57.3 \text{ kJ mol}^{-1}$). [2]

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Question 4: Enthalpy of Solution of Anhydrous vs Hydrated Sodium Carbonate

Cambridge Syllabus Reference: 9701/32/O/N/22/Q2 · [10 Marks]

Data:

- Dissolving 5.30 g anhydrous Na_2CO_3 in 50.0 cm^3 water causes temperature to rise by 4.8°C (exothermic).
- Dissolving 14.30 g hydrated $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ in 50.0 cm^3 water causes temperature to drop by 4.2°C (endothermic).

(a) Calculate the molar enthalpy of solution of anhydrous Na_2CO_3 . [$M_r = 106.0$] [3]

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(b) Calculate the molar enthalpy of solution of hydrated $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. [$M_r = 286.0$] [3]

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(c) Construct an energy cycle (Hess's Law) and calculate the enthalpy of hydration for: $\text{Na}_2\text{CO}_3(\text{s}) + 10\text{H}_2\text{O}(\text{l}) \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}(\text{s})$. [4]

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Question 5: Calorimetric Measurement of Endothermic Dissolution: Ammonium Nitrate

Cambridge Syllabus Reference: 9701/35/M/J/23/Q2 · [10 Marks]

Method: 8.00 g of ammonium nitrate, NH_4NO_3 ($M_r = 80.0$), is dissolved in 50.0 cm^3 of water in an insulated cup. The initial temperature was 21.0°C and the minimum temperature recorded was 9.5°C .

(a) Calculate the temperature drop and the heat energy absorbed, q . [2]

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(b) Calculate the molar enthalpy of solution, ΔH_{sol} , of NH_4NO_3 in kJ mol^{-1} . [3]

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(c) Explain in terms of lattice energy and hydration enthalpies why the dissolution of NH_4NO_3 is endothermic. [3]

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(d) Suggest how this reaction is utilized in emergency medicine. [2]

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Question 6: Indirect Determination of Enthalpy of Decomposition of Magnesium Carbonate via Hess's Law

Cambridge Syllabus Reference: 9701/34/M/J/22/Q2 · [10 Marks]

It is impossible to measure directly the enthalpy change for: $\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$.
A student determines this indirectly by measuring the enthalpy changes of reaction of MgCO_3 and MgO with excess $\text{HCl}(\text{aq})$.

(a) Reaction 1: $\text{MgCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ gives $\Delta H_1 = -52.0 \text{ kJ mol}^{-1}$.

Reaction 2: $\text{MgO}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$ gives $\Delta H_2 = -151.0 \text{ kJ mol}^{-1}$.

Construct the Hess's Law cycle. [4]

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(b) Calculate ΔH for the thermal decomposition of MgCO_3 . [2]

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(c) State why this decomposition cannot be measured directly in a calorimeter. [2]

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(d) Predict whether calcium carbonate requires a higher or lower decomposition temperature than MgCO_3 . Explain. [2]

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Question 7: Limiting Reactant Evaluation in Metal–Acid Displacement Calorimetry**Cambridge Syllabus Reference: 9701/31/O/N/21/Q2 · [10 Marks]**

1.20 g of magnesium ribbon is added to 40.0 cm³ of 1.50 mol dm⁻³ HCl in a polystyrene cup.

Equation: $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

Initial temperature = 19.5°C; Maximum temperature = 47.0°C.

(a) Calculate the moles of Mg and moles of HCl present, and identify the limiting reactant. [Ar: Mg = 24.3] [3]

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(b) Calculate the heat evolved, q , in the reaction. [2]

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(c) Calculate the molar enthalpy change per mole of magnesium reacted. [3]

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(d) State the hazard associated with hydrogen gas evolution during this vigorous reaction. [2]

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Question 8: Systematic Evaluation of Heat Loss in Solution Calorimetry and Apparatus Improvements

Cambridge Syllabus Reference: 9701/32/M/J/23/Q2 · [10 Marks]

A student determines the enthalpy change of a reaction using a simple polystyrene cup with no lid. The experimental value obtained is $-42.5 \text{ kJ mol}^{-1}$, compared to the accepted data book value of $-54.0 \text{ kJ mol}^{-1}$.

(a) Calculate the percentage experimental discrepancy between the student's result and the data book value. [2]

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(b) Identify two primary avenues of heat loss in this experiment. [2]

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(c) Suggest three specific apparatus modifications to significantly reduce heat loss. [3]

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(d) Explain why assuming the specific heat capacity of the solution is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ introduces a small systematic error. [3]

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Question 9: Analysis of Thermometer Uncertainty and Temperature Increment Sensitivity

Cambridge Syllabus Reference: 9701/33/M/J/21/Q2 · [10 Marks]

Thermometer A is graduated in 1°C divisions (uncertainty $\pm 0.5^{\circ}\text{C}$ per reading).
Thermometer B is graduated in 0.1°C divisions (uncertainty $\pm 0.05^{\circ}\text{C}$ per reading).
Two experiments are compared: Experiment 1 has $\Delta T = 3.5^{\circ}\text{C}$; Experiment 2 has $\Delta T = 25.0^{\circ}\text{C}$.

(a) Calculate the percentage uncertainty in measuring ΔT in Experiment 1 using Thermometer A. [3]

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(b) Calculate the percentage uncertainty in measuring ΔT in Experiment 2 using Thermometer A. [2]

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(c) Explain why Experiment 1 with Thermometer A is unacceptable for quantitative examination purposes. [2]

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(d) Suggest two ways to increase ΔT in a solution calorimetry experiment to minimize percentage uncertainty. [3]

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Question 10: Graphical Analysis of Cooling Curves and Identification of Experimental Anomalies

Cambridge Syllabus Reference: 9701/34/M/J/24/Q2 · [10 Marks]

A student records the cooling curve for an exothermic displacement reaction. At $t = 4.5$ min, the student reads 36.5°C , while readings at 4.0 min and 5.0 min are 41.0°C and 40.5°C respectively.

(a) Identify the anomalous reading and suggest a plausible laboratory error that caused it. [3]

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(b) Describe how an anomalous point should be treated when drawing the line of best fit. [2]

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(c) State the mathematical condition for drawing two straight lines of best fit to determine ΔT . [3]

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(d) Explain why the vertical distance between the two lines at $t = 3.0$ min represents the true corrected temperature rise. [2]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical: Topic 5 — Chemical Energetics

Question 1: Enthalpy of Displacement of Copper(II) by Zinc with Cooling Curve Extrapolation [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table completed with consistent temperatures to 0.5°C [1]; correct trends before/after mixing [2].	[3]
(b)	To account for heat loss to the surroundings that occurs while the reaction is taking place before peak temperature is reached [2].	[2]
(c)	$q = mc(\Delta T) = 50.0 \times 4.18 \times 24.5 = 5120.5 \text{ J} = 5.12 \text{ kJ}$ [2].	[2]
(d)	Moles $\text{CuSO}_4 = 0.800 \times 0.0500 = 0.0400 \text{ mol}$ [1]; $\Delta H = -5.12 / 0.0400 = -128 \text{ kJ mol}^{-1}$ [2] (negative sign required).	[3]

Question 2: Determination of Enthalpy Change of Neutralization (HCl + NaOH) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Total mass = $50.0 + 50.0 = 100.0 \text{ g}$ [1]; $\Delta T = 33.8 - 20.5 = 13.3^\circ\text{C}$ [1]; $q = 100.0 \times 4.18 \times 13.3 = 5559.4 \text{ J} = 5.56 \text{ kJ}$ [1].	[3]
(b)	Moles = $2.00 \times 0.0500 = 0.100 \text{ mol}$ [2].	[2]
(c)	$\Delta H = -5.56 / 0.100 = -55.6 \text{ kJ mol}^{-1}$ [2].	[2]
(d)	Strong acids and bases fully dissociate in aqueous solution; the reaction in all cases is simply $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ with spectator ions unchanged [3].	[3]

Question 3: Comparison of Neutralization Enthalpies: Ethanoic Acid vs Hydrochloric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Ethanoic acid is a weak acid and is only partially dissociated in aqueous solution [1]; energy is absorbed (endothermic process) to break O-H bonds to fully ionize the remaining un-ionized acid molecules [2]; this absorbed energy reduces the overall exothermic heat output [1].	[4]
(b)	$\text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [2].	[2]
(c)	Glass is a much better thermal conductor than expanded polystyrene, leading to higher rates of heat loss to surroundings and the beaker itself absorbs substantial heat [2].	[2]
(d)	% diff = $[(57.3 - 50.6) / 57.3] \times 100 = 11.69\%$ [2].	[2]

Question 4: Enthalpy of Solution of Anhydrous vs Hydrated Sodium Carbonate [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	$q = 50.0 \times 4.18 \times 4.8 = 1003.2 \text{ J} = 1.003 \text{ kJ}$ [1]; Moles = $5.30 / 106.0 = 0.0500 \text{ mol}$ [1]; $\Delta H = -1.003 / 0.0500 = -20.1 \text{ kJ mol}^{-1}$ [1].	[3]
(b)	$q = 50.0 \times 4.18 \times (-4.2) = -877.8 \text{ J} = -0.878 \text{ kJ}$ [1]; Moles = $14.30 / 286.0 = 0.0500 \text{ mol}$ [1]; $\Delta H = +0.878 / 0.0500 = +17.6 \text{ kJ mol}^{-1}$ [1].	[3]
(c)	$\Delta H_{\text{hydration}} = \Delta H_{\text{sol}}(\text{anhydrous}) - \Delta H_{\text{sol}}(\text{hydrated})$ [2]; $\Delta H_{\text{hydration}} = -20.1 - (+17.6) = -37.7 \text{ kJ mol}^{-1}$ [2].	[4]

Question 5: Calorimetric Measurement of Endothermic Dissolution: Ammonium Nitrate [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	$\Delta T = 9.5 - 21.0 = -11.5^\circ\text{C}$ [1]; $q = 50.0 \times 4.18 \times 11.5 = 2403.5 \text{ J} = 2.404 \text{ kJ}$ [1].	[2]
(b)	Moles = $8.00 / 80.0 = 0.100 \text{ mol}$ [1]; $\Delta H = +2.404 / 0.100 = +24.0 \text{ kJ mol}^{-1}$ [2] (positive sign required).	[3]

(c)	$\Delta H_{\text{sol}} = \text{Lattice Energy} + \Delta H_{\text{hydration}}$ [1]; the magnitude of the endothermic lattice energy required to separate ions is greater than the exothermic hydration energy released when ions are solvated by water molecules [2].	[3]
(d)	Used in instant cold packs to treat sporting injuries, sprains, and reduce swelling without refrigeration [2].	[2]

Question 6: Indirect Determination of Enthalpy of Decomposition of Magnesium Carbonate via Hess's Law [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Cycle drawn showing MgCO_3 and MgO both reacting with 2HCl to form $\text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [2]; $\Delta H_{\text{target}} = \Delta H_1 - \Delta H_2$ [2].	[4]
(b)	$\Delta H = -52.0 - (-151.0) = +99.0 \text{ kJ mol}^{-1}$ [2].	[2]
(c)	Thermal decomposition requires high temperatures (Bunsen flame) so heat is constantly supplied, preventing simple calorimetric measurement [2].	[2]
(d)	Higher temperature [1]; Ca^{2+} has a larger ionic radius and lower charge density than Mg^{2+} , causing less polarization of the carbonate ion [1].	[2]

Question 7: Limiting Reactant Evaluation in Metal-Acid Displacement Calorimetry [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles $\text{Mg} = 1.20 / 24.3 = 0.04938 \text{ mol}$ [1]; Moles $\text{HCl} = 1.50 \times 0.0400 = 0.0600 \text{ mol}$ [1]; 0.0600 mol HCl reacts with $0.0300 \text{ mol Mg} \rightarrow \text{HCl}$ is the limiting reactant [1].	[3]
(b)	$\Delta T = 47.0 - 19.5 = 27.5^\circ\text{C}$ [1]; $q = 40.0 \times 4.18 \times 27.5 = 4598 \text{ J} = 4.60 \text{ kJ}$ [1].	[2]
(c)	Moles of Mg reacted = moles of $\text{HCl} / 2 = 0.0600 / 2 = 0.0300 \text{ mol}$ [1]; $\Delta H = -4.60 / 0.0300 = -153.3 \text{ kJ mol}^{-1}$ [2].	[3]
(d)	Hydrogen is highly flammable and forms explosive mixtures with air; keep away from open flames [2].	[2]

Question 8: Systematic Evaluation of Heat Loss in Solution Calorimetry and Apparatus Improvements [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	% error = $[(54.0 - 42.5) / 54.0] \times 100 = 21.3\%$ [2].	[2]
(b)	Convection and evaporation from the unsealed top surface [1]; conduction through cup walls and to the thermometer/stirrer [1].	[2]
(c)	1. Add a fitted polystyrene lid with a small hole for the thermometer [1]; 2. Double-cup the calorimeter / nest inside a second polystyrene cup inside a glass beaker [1]; 3. Use a vacuum Dewar flask instead of polystyrene [1].	[3]
(d)	4.18 is the heat capacity of pure water; dissolved solute ions alter bonding and heat capacity, typically lowering the solution's actual heat capacity slightly below 4.18 [3].	[3]

Question 9: Analysis of Thermometer Uncertainty and Temperature Increment Sensitivity [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Two temperature readings (initial and final) $\rightarrow$ total uncertainty = $2 \times 0.5 = 1.0^\circ\text{C}$ [1]; % uncertainty = $(1.0 / 3.5) \times 100 = 28.6\%$ [2].	[3]
(b)	% uncertainty = $(1.0 / 25.0) \times 100 = 4.0\%$ [2].	[2]
(c)	A 28.6% apparatus uncertainty exceeds acceptable scientific error margins ($<5\%$) and renders the calculated enthalpy change unreliable [2].	[2]
(d)	Increase the concentration of the reactants [1.5]; decrease the volume of solvent/water while maintaining the same moles of solute [1.5].	[3]

Question 10: Graphical Analysis of Cooling Curves and Identification of Experimental Anomalies [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Reading at $t = 4.5$ min (36.5°C) is anomalous [1]; Cause: misreading the thermometer scale by 5°C (read 36.5 instead of 41.5) or thermometer bulb temporarily lifted out of liquid or incomplete initial stirring [2].	[3]
(b)	Circle or clearly identify the point as an anomaly and ignore it when plotting the cooling line of best fit [2].	[2]
(c)	Line 1: Extrapolate initial pre-mixing temperatures forward to the time of mixing ($t = 3.0$ min) [1.5]; Line 2: Extrapolate post-peak cooling temperatures backward to $t = 3.0$ min [1.5].	[3]
(d)	It models the hypothetical instantaneous reaction temperature rise assuming zero heat was lost to the environment during reaction [2].	[2]